

課程資訊	
課程名稱	適應控制 ADAPTIVE CONTROL
開課學期	95-2
授課對象	工學院 機械工程學研究所
授課教師	陽毅平
課號	ME7131
課程識別碼	522EM3210
班次	
學分	3
全/半年	半年
必/選修	選修
上課時間	星期三2,3,4(9:10~12:10)
上課地點	工綜213
備註	本課程以英語授課。 總人數上限：70人
Ceiba 課程網頁	http://ceiba.ntu.edu.tw/952adaptivecontrol
課程簡介影片	
核心能力關聯	核心能力與課程規劃關聯圖

課程大綱

為確保您的權利,請尊重智慧財產權及不得非法影印

課程概述	Course Description: Requisite: Automatic Control, and Digital Control. This course provides a broad introduction to adaptive control techniques and practical implementations. This course covers adaptive control theories for deterministic and stochastic* systems, including recursive system identification, self-tuning regulator, model-reference adaptive system, auto-tuning, gain scheduling*, and adaptive control with lattice filter. (*optional)
課程目標	Course Objectives: The primary goal of this introductory course is to help students become knowledgeable designers of adaptive control systems. A moderate amount of simulations and experiments are associated with Matlab/Simulink toolboxes.
課程要求	Course Grading: Homework 40% Homework is normally due in one week after assignment, A4 papers Term Project 30% Research-related or industry-related practical issues Paper Reading 20% 10 Journal Papers, each with a 2-page report Attendance 10%

預期每週課前或/與課後學習時數																	
Office Hours																	
指定閱讀																	
參考書目	Textbooks: (1) K.J. and B. Wittenmark, Adaptive Control, 2nd ed., Addison-Wesley Publishing Company, 1995. (2) G.C. Goodwin and K. S. Sin, Adaptive Filtering, Prediction and Control, Prentice-Hall, Inc., Englewood Cliffs, New Jersey, 1984. References: [1] B. Widrow and E. Walach, Adaptive Inverse Control, Prentice Hall, 1996. [2] Y. D. Landau, Adaptive Control - The Model Reference Approach, New York, Marcel Dekker, 1979. [3] L. Ljung, System Identification: Theory for The User, Prentice-Hall, Inc., 1987. [4] K. S. Narendra and A. M. Annaswamy, Stable Adaptive Systems, Prentice-Hall International Ed., 1989. [5] S. Sastry and M. Bodson, Adaptive Control - Stability, Convergence, and Robustness, Prentice-Hall International Ed. 1989. [6] B. D. O. Aderson, et al., Stability of Adaptive Systems --- Passivity and Averaging Analysis, Cambridge, Mass.: MIT Press, 1986. [7] T. and P. Stoica, System Identification, Prentice-Hall, Inc., 1989. [8] B. Roffel, P. J. Vermeer and P. A. Chin, Simulation and Implementation of Self-Tuning Controllers, Prentice-Hall, Inc., 1989. [9] E. Mosca, Optimal, Prediction, and Adaptive Control, Prentice-Hall, Inc., 1995. [10] Every book and journal paper about adaptive control.																
評量方式 (僅供參考)	<table><tr><th>No.</th><th>項目</th><th>百分比</th><th>說明</th></tr><tr><td>1.</td><td>Homework</td><td>40%</td><td>Homework is normally due in one week after assignment, A4 papers</td></tr><tr><td>2.</td><td>Term project</td><td>30%</td><td>Research-related or industry-related practical issues</td></tr><tr><td colspan="4"></td></tr></table>	No.	項目	百分比	說明	1.	Homework	40%	Homework is normally due in one week after assignment, A4 papers	2.	Term project	30%	Research-related or industry-related practical issues				
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3.	paper reading	20%	10 Journal Papers, each with a 2-page report
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1. 本校尚無訂定 A+ 比例上限。
2. 本校採用等第制評定成績，學生成績評量辦法中的百分制分數區間與單科成績對照表僅供參考，授課教師可依等第定義調整分數區間。詳見學習評量專區 ([連結](#))。

課程進度

週次	日期	單元主題
第1週	2/28	holiday
第2週	3/07	What is Adaptive Control? (A&W Ch. 1)
第3週	3/14	Parameter Estimation for Deterministic Systems (A&W Ch. 2, G&S Ch. 3)
第4週	3/21	Parameter Estimation for Deterministic Systems (A&W Ch. 2, G&S Ch. 3)
第5週	3/28	Parameter Estimation for Deterministic Systems (A&W Ch. 2, G&S Ch. 3)
第6週	4/04	Adaptive Control of Linear Deterministic Systems (A&W Ch. 3, G&S Ch. 6)
第7週	4/11	Adaptive Control of Linear Deterministic Systems (A&W Ch. 3, G&S Ch. 6)
第8週	4/18	Adaptive Control of Linear Deterministic Systems (A&W Ch. 3, G&S Ch. 6)
第9週	4/25	Model Reference Adaptive Control (A&W Ch. 5, Landau)
第10週	5/02	Model Reference Adaptive Control (A&W Ch. 5, Landau)
第11週	5/09	Model Reference Adaptive Control (A&W Ch. 5, Landau)
第12週	5/16	Auto-Tuning (A&W Ch. 8)
第13週	5/23	Adaptive Control with Lattice Filter -- Order Update (G&S Ch. 9)
第14週	5/30	Applications (Paper review)
第15週	6/06	Applications (Paper review)
第16週	6/13	Term Project